

S019A Findings: What Works Without Calibration Fix

Finding	Evidence	Paper Claim Supported
Sensitivity gradient (IG-3) HEADLINE	eta-sq(embedding): 0.549 (diabetes) < 0.618 (pop density) < 0.687 (elevation) All $p < 0.001$; monotonic increase confirmed	Embedding sensitivity scales with task difficulty -- not binary "works/doesn't" Stronger claim than original IG-1 invariance hyp.
Embedding effect (IG-1 revised + IG-2)	ANOVA F: 39.6 (diabetes), 130.7 (pop), 197.8 (elevation); all $p < 0.001$	Architecture choice matters for ALL geo-tasks, but magnitude depends on problem regime
Theory kappa discriminates	Spearman(R-sq, theory kappa) = 0.482 overall ($p < 0.001$); per-target: 0.23--0.44	D*/D correctly ranks embedding quality even when RSN simplex is degenerate
N-ceiling pattern	N_ceil: 0.220 (diabetes), 0.187 (pop), 0.281 (elevation) -- varies by target	Task difficulty is a first-class evaluation axis, measurable from solver residuals
Solver robustness	Solver eta-sq: 0.141--0.216 across targets; interaction eta-sq: 0.035--0.091 (small)	Embedding ranking is stable across solver families -- not an artifact of model choice

Known limitation: RSN simplex, alpha, gate decisions are degenerate (per-arm tercile calibration -- see Appendix F).
Theory kappa (D*/D) and R-sq are unaffected. IG-4 (gate divergence) and IG-5 (simplex integrity) require shared-reference recalibration.